

Roll No.....

END SEMESTER EXAMINATION DEC - 2025

Name of the Program: B. Tech Petroleum Engineering

Semester: 3rd

Name of the Course: Petroleum Drilling Engineering I

Course Code: TPE 304

Time: 3 Hrs

Maximum Marks: 100

Notes:

- (i) All questions are compulsory
- (ii) Answer any two sub questions among a,b,c in each main question.
- (iii) Total marks for each main question is twenty. Each questions carries 10 marks

Q1	(2X10 = 20 marks)	CO
(a)	Explain the purpose and components of a drilling rig hoisting system. Draw a labeled diagram and describe the role of each component.	CO1
(b)	What is a top drive system in drilling operations? How does a top drive differ from a conventional rotary table and Kelly drive system? What are the advantages of using a top drive compared to a Kelly drive?	CO1
(c)	A drillstring with a buoyant weight of 200,000 lbs must be pulled from the well. A total of 8 lines are strung between the crown block and the travelling block. Assuming that a four sheave, roller bearing system is being used. i. Compute the tension in the fast line ii. Compute the tension in the deadline iii. Compute the vertical load on the rig when pulling the string.	CO2
Q2	(2X10 = 20 marks)	CO
(a)	What is drilling fluid? What are different types of drilling fluid? List the main functions of drilling fluid?	CO3
(b)	A well is being drilled to a true vertical depth (TVD) of 10,000 ft using a drilling mud with a density of 10 ppg (pounds per gallon). Calculate the hydrostatic pressure at the bottom of the well in psi. If the formation pressure is 4,600 psi, determine the overbalance or underbalance condition and by how much.	CO2
(c)	List the properties of drilling fluid. Discuss the importance of properties such as density, viscosity, gel strength, and filtration.	CO3
Q3	(2X10 = 20 marks)	CO
(a)	What are the the purpose of casing in oil well drilling operation? Describe the main types of casing and their functions.	CO4
(b)	What are the functions of cementing operations in oil well? What are primary, liner, squeeze and plug back cementing and when is it used? Explain	CO4
(c)	Discuss how cement bond logs (CBL) are used to evaluate cement integrity.	CO4

Q4	(2X10 = 20 marks)	CO
(a)	Explain the components and working of a well control system used in drilling operations. Include the role of blowout preventers (BOPs), choke manifold, and mud pumps in maintaining well control. Draw a labeled schematic diagram.	CO5
(b)	Explain the concept of hydrostatic pressure in a fluid column. Discuss how fluid density and true vertical depth (TVD) influence the pressure at the bottom of a well.	CO5
(c)	Define Reynolds number and explain its significance in determining flow regimes. Why is turbulent flow often desirable in drilling operations? Explain how drilling fluid properties (density and viscosity) affect whether the flow becomes turbulent.	CO5
Q5	(2X10 = 20 marks)	
(a)	List the two main types of bits in use today. Also list two subclassifications of each basic bit type and discuss the conditions considered ideal for the application of each subclassification given.	CO6
(b)	List seven rules of thumb of bit selection. What is the best basis of comparison when trying to choose between two different bit types?	CO6
(c)	Discuss the factors affecting penetration of a drill bit?	CO6